9/7/25 — COMPUTER ORGANIZATION & ARCHITECTURE —

Topics — computer fundamental

- No. system [binary, octal, decimal & hexa-decimal no.
- binary no. representation : sine magnitude 1's & 2's complement representation.
- logic gates : [basic, universal, derived] AND, OR, NOT NAND, NOR, XOR, - Basic organization of computer, BOA

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- Binary

Decimal	Hexa	8	4	2	1
0	0	0	0	0	0
1	1	0	0	0	1
2	2	0	0	1	0
3	3	0	0	1	1
4	4	0	1	0	0
5	5	0	1	0	1
6	6	0	1	1	0
7	7	0	1	1	1
8	8	1	0	0	0
9	9	1	0	0	1
10	A	1	0	1	0
11	B	1	0	1	1
12	C	1	1	0	0
13	D	1	1	0	1
14	E	1	1	1	0
15	f	1	1	1	1

* Binary to decimal conversion

1. $(1101)_2$

$1\times2^3+1\times2^2+0\times2^1+1\times2^0$

$=8+4+0+1$

$=(13)_2$

2. $(110110)_2$

$1\times2^5+1\times2^4+0\times2^3+1\times2^2+1\times2^1+0\times2^0$

$=(54)_2$

3. $(11011.110)_2$

$1\times2^4+1\times2^3+0\times2^2+1\times2^1+1\times2^0 . 1\times2^{-1}+1\times2^{-2}+1\times2^{-3}$

$=16+8+0+2+1$ $\quad =0.75$

$=24+2+1$

$=27$ $\quad \therefore (27.75)_{10}$.

4. convert $(12)_{10}$ into its binary equivalent.

$\frac{12}{2}=0$

$6/2=0$

$3/2=1$

$1=1$

$\therefore (12)_{10}=(1100)_2$

5. convert $(82)_{10}$ into its binary equivalent.

$82/2=0$

$41/2=1$

$20/2=0$

$10/2=0$

$5/2=1$

$2/1=0$

$1=1$

$\therefore (82)_{10}=(1010010)_2$

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6. $(27.75)_{10}$ into it's binary equivalent

$27/2 = 1$

$13/2 = 1$

$6/2 = 0$

$3/2 = 1$

$1 = 1$

$\therefore 0.75 \times 2 = 1$

$0.75 \times 2 = 1$

$\therefore (27.75)_{10} = (11011.11)_2$

7. Convert $(951.23)_{10}$ into binary equivalent

$951/2 = 1$ $\quad 0.23 \times 2 = 0$

$475/2 = 1$ $\quad 0.46 \times 2 = 0$

$237/2 = 1$ $\quad 0.92 \times 2 = 1$ $\quad \therefore (951.23)_{10} = (1110110111.0011100)_2$

$118/2 = 0$ $\quad 0.84 \times 2 = 1$

$59/2 = 1$ $\quad 0.68 \times 2 = 1$

$29/2 = 1$ $\quad 0.36 \times 2 = 0$

$14/2 = 0$ $\quad 0.72 \times 2 = 1$

$7/2 = 1$ $\quad 0.44 \times 2 = 0$

$3/2 = 1$ $\quad 0.88 \times 2 = 1$

$1 = 1$ $\quad 0.76 \times 2 =$

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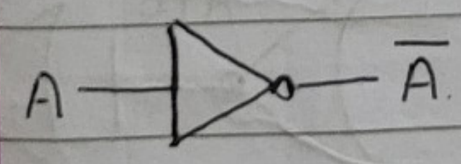
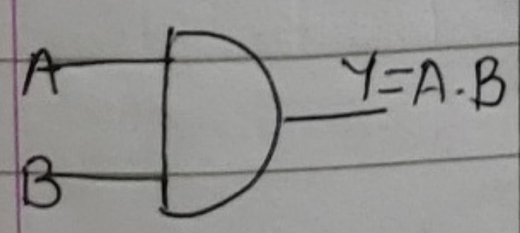
- Basic Gates.

1. NOT Gate.



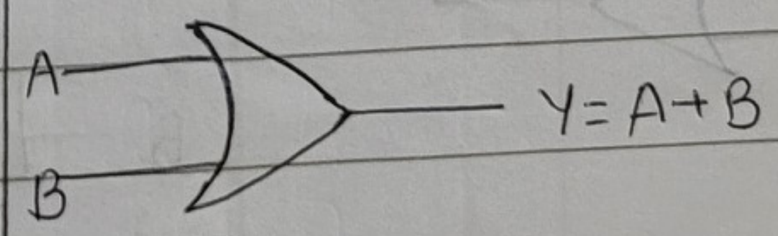
A	$\bar{A}$
0	1
1	0

2. AND Gate.



A	B	Y=A.B
0	0	0
0	1	0
1	0	0
1	1	1

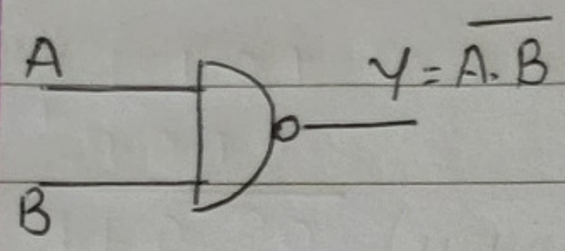
3. OR Gate.



A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

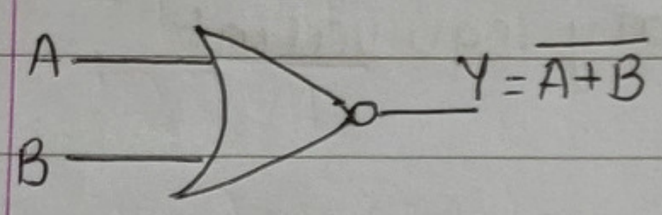
- Universal Gates.

1. NAND Gate.



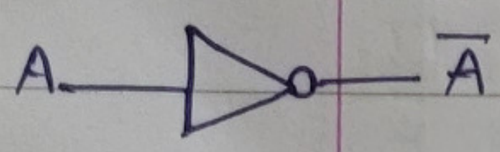
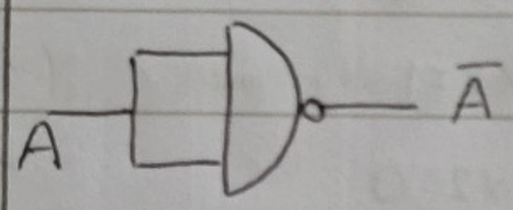
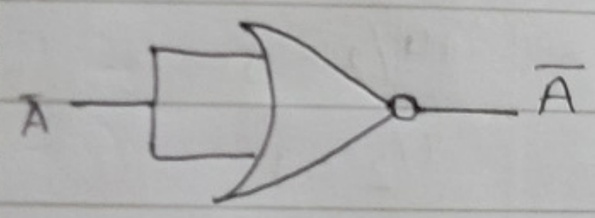
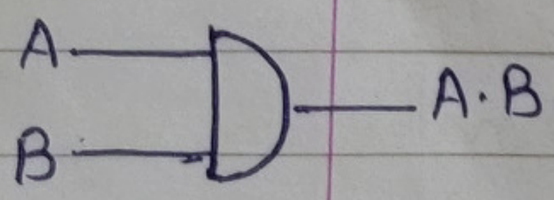
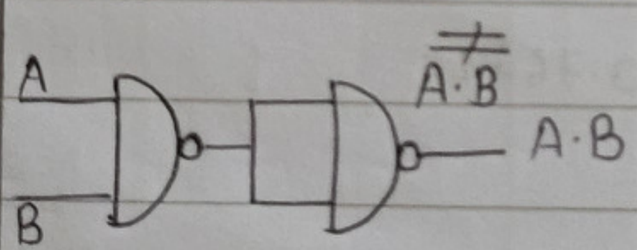
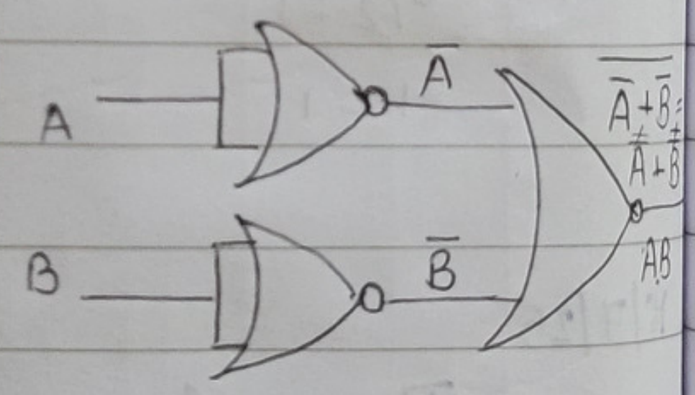
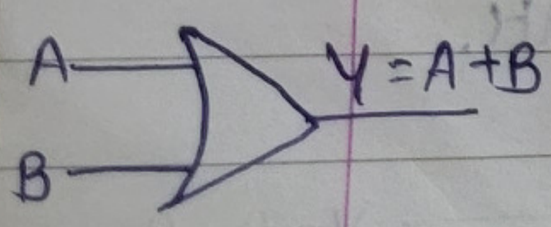
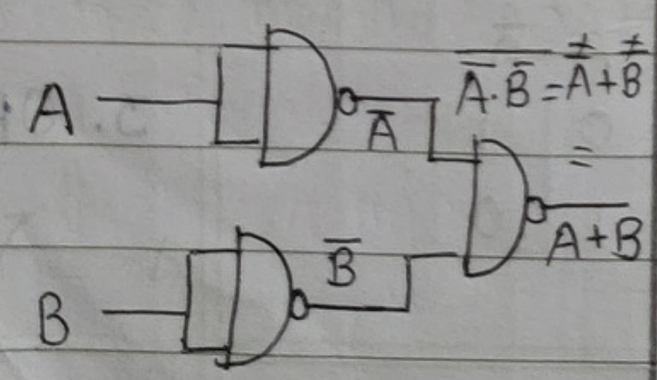
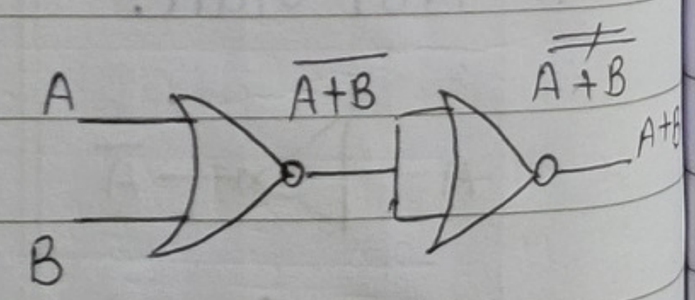
A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

2. NOR Gate.



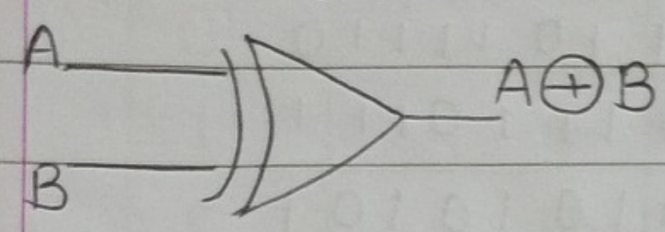
A	B	$\overline{A+B}$
0	0	1
0	1	1
1	0	1
1	1	0

- Implementation of Basic gate using universal gate

	NAND	NOR.
1. NOT 		
2. AND 		
3. OR 		

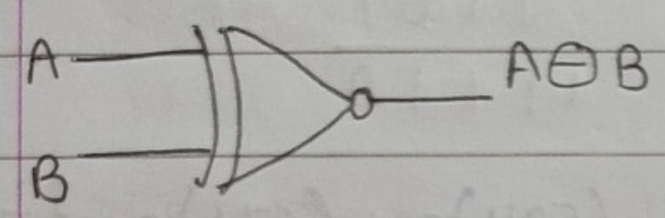
- Derived Gate

1. Ex-OR gate.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

2. Ex-NOR gate.



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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- 1's complement
  0 - 1
  1 - 0

- 2's complement [add 1 to last]

  Eg:- 1 0 1 1 0 1 1
  0 1 0 0 1 0 0 — 1's complement
  0 1 0 0 1 0 1 — 2's complement

- Sine magnitude.
  check 1st (MSB) if 1 then it's Positive
  and if 0 then it is negative.